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An embedded system approach for energy monitoring and analysis in industrial processes

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ABSTRACT

This paper presents the development of a low cost embedded system targeted to energy management in industrial environments. The main objective of this study is a methodology for monitoring, acquisition and analysis of electrical parameters using digital signal processing (DSP) techniques. For this task, a system (hardware and software) responsible for the acquisition records the input signals using low cost technologies, such as microcontrollers, DSP platforms and embedded system like ARM System-on-Chip (SoC) Architecture. Finally, for data analysis, two applications were created: one using the Matlab® software, and another derived from the first, to run on the embedded device. Both applications used Discrete Wavelet Transform as primary technique of disturbances detection. As a result, it is shown a complete system for monitoring and analysis of electrical parameters, serving as a tool to aid the energy management in industrial processes.

1. Introduction

Electrical Power Quality (EPQ) has become a growing and common concern to electricity companies (state concessions, electrification cooperatives) and consumers in general [17,28]. This growing interest in EPQ is due mainly to the technological evolution of electronic equipments, widely used in various sectors of activity (industrial, commercial and residential). This development resulted in more sensitive to electrical disturbances equipments [10]. These disturbances are mainly generated by the application of power electronics in several industrial processes, such as those involving the automation and energy conversion (AC/DC - Alternating Current to Direct Current).

The study of power quality as well as improvements in Energy Efficiency (EE) in electrical systems encompasses the analysis, diagnosis, and the proposition of possible solutions for the identified problems and the technical-economic evaluation [17]. For Musolino et al. [20] high efficiency energy storage systems permit energy recovery, peak shaving and power quality functions. Energy efficiency contributes significantly in reduce energy security risks and emission problems [11]. According to Garcia et al. [10], previous studies in energy generation and consumption field have shown that technical, commercial and financial losses sum up costs totaling billions of dollars a year worldwide, and its main cause is associated to the EPQ. Thus, one should establish joint actions between electric utilities, consumers and equipment manufacturers. For Hackl and Harvey [12] it is important to involve all companies within the cluster in studies to a more resource efficient and competitive production. Therefore, the problems of power quality are diagnosed and solved early in the projects elaboration. However, it is of great importance that corrective actions and awareness programs for the wise use of electricity should be applied and maintained permanently.

Thus, actions are necessary to establish energy efficiency, once they represent socioeconomic activities that aim to provide better power consumption (hydraulic, electric, gas, fuel). Therefore, seeking for alternatives and improving the efficiency of industrial processes are for industrial managers, rather than a sense of necessity, a matter of survival in the market [10,26,27]. The sustainable development of an industry is directly connected with energy efficiency [23]. The economic energy efficiency is a factor which is linking with other macroeconomic variables [31]. The reduction of energy consumption, besides being important to preserve the natural environment, as it reduces the need to increase supply, also provides financial returns for companies.

It becomes important that industries have effective management of energy demand consumed daily, so avoiding energy waste as a decrease in quality. However, given that the EPQ can reach the consumer with an excellent standard of quality, but with incorrect

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management, or even problems arising from the own infrastructure of the company, may imply in the reduction in the quality of the energy consumed in the industrial process. In this context, the focus of this work is the development of an embedded system (ES - hardware and software) capable of capturing, storing and analyzing information related to consumption and power quality in industrial systems. Furthermore, a method to evaluate the applied EPQ parameters is proposed, mapping in detail the power consumption in an industry (organization being researched), identifying the sectors of greatest consumption, detect and classify disturbances, searching for possible solutions.

2. EPQ disorders detection using the wavelet transform approach

According to Solórzano [28] the EPQ refers to a wide variety of conducted electromagnetic phenomena that characterize the voltage and current at a given time and location of the electrical system, directly affecting the energy efficiency of industrial processes.

Thus, the EPQ has three quality levels, namely: quality of customer service, service quality and product quality. The quality of customer service refers to the unauthorized charges, fees, service time and more. The quality of service is related to the operation and maintenance of the electrical system, providing to customers the minimal disruption acceptable. As for product quality it has a technical focus, regarding the compliance of the product itself, the availability of electric power with balanced sinusoidal voltages and with constant amplitude and frequency [4,8,10].

According to Deckmann and Pomilio [3] it is necessary for technicians or experts a research to diagnose the causes of the problems related to power quality. As it comes to diagnosing a problem of electromagnetic compatibility or the search for indicators of electrical parameters outside standards, this research may involve issues that go beyond a simple technological problem.

The EPQ refers to a wide variety of conducted electromagnetic phenomena that characterize the voltage and current at a given time and location of the electrical system. Thus, the quality of energy in a particular region of the electrical system is adversely affected by a wide variety of disorders [3,8,28], as: (i) transients (impulsive and oscillatory); (ii) short duration variations (transitory outages, voltage sags and voltage jumps); (iii) long duration variations (sustained interruptions, undervoltage and overvoltage); (iv) waveform distortion (voltage cut-off, harmonics, noise); and (v) voltage fluctuations and frequency variations.

For Ferreira [7] the identification and classification of power disturbances is an important step in the process of monitoring the EPQ, since it can directly contribute to identify the causes of the disturbances. Now the classification should be preceded by the step of detecting disorders. Various techniques can be used, as the average value, the Discrete Wavelet Transform (DWT), the Fourier Transform (FT) or Fast Fourier Transform (FFT). Even computational intelligence has been applied in the detection of disorders in EPQ [5,24,25]. Thus, functions or algorithms which permit the extraction of its characteristics are necessary to identify a disorder [10]. This way, the following methods are explained for identifying disorders: (i) RMS value calculation; (ii) Instantaneous Euclidean Norm; (iii) Application of Discrete Fourier Transform; and (iv) Application of Wavelet Transform.

The Wavelet Transform has some peculiarities that make it suitable for EPQ event detection in EPQ. One of these peculiarities is the behavior of these transform coefficients when the signal presents singularities. It is noteworthy that singularities in the case of EPQ can be understood as the moment of an event occurrence (switching capacitors, voltage sags, voltage interruption, etc.). When the presence of singularity, the coefficients tend to generate local peaks in each of the scales. Connecting these peaks at different scales, we obtain a line, known as *maxima line*. This line converges to the instant of the singularity occurrence.

In other words, the Wavelet Transform is used when it is necessary to locate individual events in the time domain [13], whereas the classical method of spectral analysis of frequency using Fourier Transform is suitable for periodic signals in steady state. This transform has its continuous (CWT) and discrete (DWT) variants. For Filho and Garcia et al. [9,10] the signal frequency components can be separated using wavelet decomposition filters. The result obtained when using the high-pass decomposition filter allows obtaining what is known as the detail coefficients in the wavelet domain, and the sequences of detail in time domain (Fig. 1).

According to Ferreira [8], the effect of the scale change of a signal can be best interpreted using the concept of resolution, this being achieved using filters. The filtering process used for the DWT presents an embodiment of the Multiresolution Analysis (MRA) technique proposed by Mallat [18], resulting in the combination of a scale function $f_{(t)}$ and a function wavelet $\psi_{(t)}$. This process is based on filtering a signal to be analyzed through high-pass and low-pass filters, providing versions of the original signal regarding the coefficients of wavelet functions and scale functions, or approximations and details, respectively [19].

The definition of a DWT function f by wavelet ψ is:

$$TWC_f^{\psi}(a,b) = \left| a \right|^{\frac{-1}{2}} \int_{-\infty}^{\infty} f(t)\psi_{ab}(t)dt \tag{1}$$

or,

$$(TWC)(a,b) = \left| a_0^m \right|^{\frac{-1}{2}} \int_{-\infty}^{\infty} f(n)\psi\left(\frac{t-b}{a}\right)dt \tag{2}$$

Thus, from the engineering point of view, DWT is no more than a digital filtering process in the time domain (discrete convolution), followed by a reduction of the number of samples (*downsampling*) by a factor of 2 [29]. Basically, the TWD function decomposes a signal into multiple frequency bands without losing information in time, thus providing an uneven division of the time-frequency plane. The same short intervals attributes in time for the high frequency components (details) and larger intervals to low frequency components (approaches). This division provides a better representation (resolution) of the signal in the areas of frequency and time [19], as shown in Fig. 2a. The One-Dimensional DWT implementation is shown in Fig. 2b.

A practical example of TWD usage can be seen in Fig. 3. The wavelet decomposition *S* (original signal in addiction to noise) results in the *cA1* and *cD1* outputs to the first level of decomposition. Applying another decomposition level for *cA1* we have the *cA2* and *cD2* signals. It is possible to observe an improvement in the quality of *cA2* signal relative to *S*[19].

3. Related works

Several works have been developed by academic and scientific communities in the context of EPQ. They typically target the development of techniques, methods and systems, that aim to monitor, analyze and measure related events. Some of these studies are based on certain mathematical tools such as Wavelet Transform and Fourier Transform [30,13]. This Section gives some background on these works and compares them.

Leborgne [17] in his dissertation presents an alternative

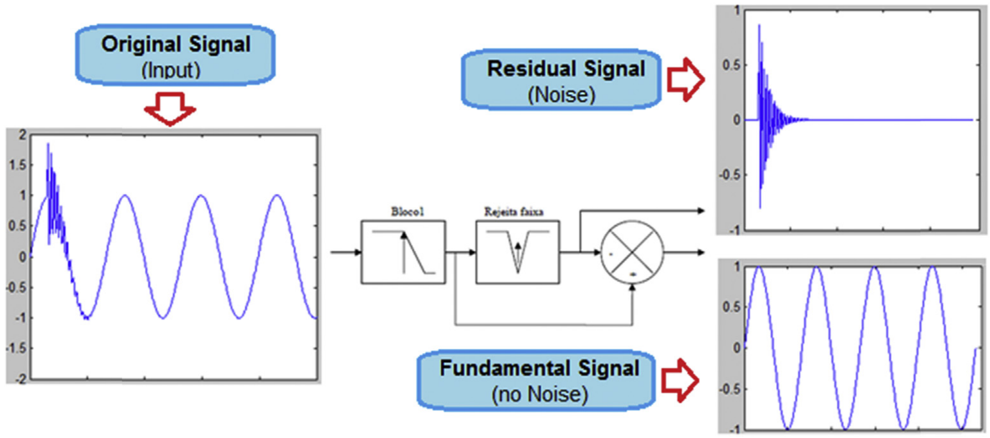

Fig. 1. Decomposition of signals with a signal disturbance. Source: Adapted from Refs. [15,16].

methodology to characterize the sensitivity of loads and industrial processes against voltage sags, using an integrated monitoring of EPQ by collecting data directly from the manufacturing process. The methodology presented in this work allows modeling the sensitivity of the loads and processes by conventional characterization (intensity and duration) as alternative methods. Oleskovicz et al. [21] carried out a comparative study between the Fourier Transform with many types of windows and Wavelet Transform with Daubechies filter [2]. After being transformed, the signals are then processed by an artificial neural network (ANN) to classify the disturbances.

Garcia et al. [10] presents a computational tool to aid in the solution of problems of EPQ, which is responsible for analyzing situations caused by possible disturbances originated in different processes. The analysis is carried out based on a monitoring system database. This system has a module for recognizing similar transient disturbances and uses MATLAB for test and validation. Fernandes et al. [6] describes the development of a technique for signals pre-processing and classification of disturbances in a given power system. Regarding the disturbance, sags, elevations, interruptions and transients were considered. Windowing was used for signals acquisition and digitization, and RNAs were used to classify occurrences. The results showed a good performance of all the proposed methodology for the signal analysis in the context of EPQ.

Junior, O. [16] presents a method for power quality analyzing and monitoring by identifying and quantifying the electromagnetic disturbances. DSP techniques were used, allowing the construction of digital filters, event detection and estimation of the frequency for the electrical signals analyzed. The software has been tested using waveforms with disorders previously known for its validation. Waveforms obtained from field measurements were analyzed, verifying the effectiveness of the algorithm against noise and other phenomena of EPQ on real world measurements. Junior, F. [15] dissertation describes the use of the Wavelet Transform and Instant Euclidean Norm for event detection and monitoring of EPQ. The system was simulated using Simulink and output data were validated in Matlab.

Finally Ferreira[7] shows in his thesis signal processing techniques and computational intelligence which have been applied to the analysis, detection and classification of power disturbances. The technique of principal lines is used for disturbances analysis. As a result, it is possible to evaluate the complexity of each class of disorders in which important parameters for the detection and classification have been revealed. In the detection, these parameters were explored in comparison with the parameters already used, such as RMS. Table 1 presents a comparison between the present work and the related ones described above. We use for this brief comparison the adoption of the following concepts: Electrical Power Quality (EPQ), Energy Efficiency (EE), Wavelet Transform (WT), Fourier Transform (FT), Embedded Systems (ES), Matlab and study case scenario.

4. Methodological procedure

To achieve the goals of creating an equipment (and a methodology) aimed to the effective management to improve energy savings, this project was divided into four stages throughout the implementation period, as follows: (i) *analysis of the existing solutions in the market, as well as the studies in the current scenario of EE* - in this stage it was also conducted the survey of theoretical reference, especially regarding the studies of Wavelet Transforms; (ii)

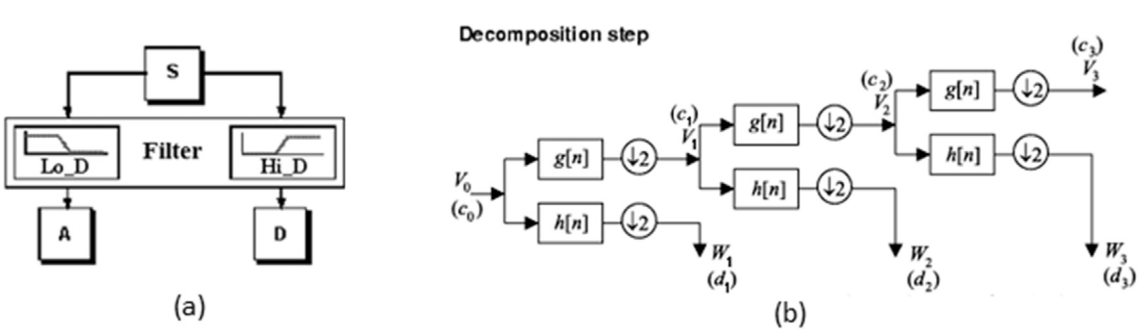

Fig. 2. Successive decompositions of DWT. Source: [19].

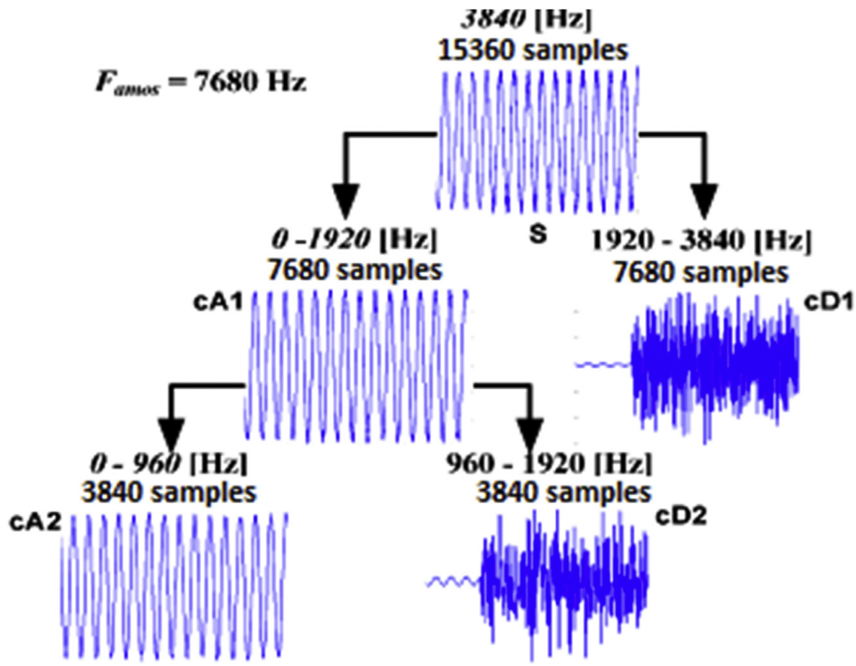

Fig. 3. Successive decomposition process of a noisy signal using DWT. Source: [29].

Table 1
— Related works comparison.

Author	EPQ	EE	WT	FT	ES	Matlab	Case
Leborgne	Y	N	N	Y	Y	Y	Y
Oleskovicz	Y	N	Y	Y	N	Y	N
Garcia	Y	N	Y	N	Y	Y	N
Fernandes	Y	N	Y	N	Y	Y	N
Junior, O.	Y	N	Y	Y	Y	Y	Y
Junior, F.	Y	N	Y	Y	Y	Y	N
Ferreira	Y	N	Y	Y	N	N	N
Back	**Y**	**Y**	**Y**	**N**	**Y**	**Y**	**Y**

Legend: Y — Yes, N — No.

implementation of, the power supply information system acquisition and analysis module - this step includes the design and dimensioning of electronic boards for signal conditioning, data acquisition and discretization and embedded systems programming; (iii) *installation and calibration of the acquisition system in an industrial environment*; and final step (iv) *analysis of the collected data* — thus, the consumption model could be raised, as well as the detection and classification of power disturbances.

Below is a description of each of the blocks presented in Fig. 4:

- Power input: represents the overall input voltage that feeds one or more industrial processes;
- Measurement: This block represents the process of data acquisition and is responsible for the analog/digital conversion;
- Event Capturing: receives data already digitized in the measurement block. It is responsible for events capturing (disturbances) for further analysis of the state of EPQ in the process;
- Consumption capturing and recording: receives data already digitized in the measurement block, being responsible for the registration of consumption;
- Model: this block owns the model which describes the energy consumption of the process, serving as the basis for the simulations, both in the matter of quantifying disturbances as determining the optimal energy efficiency of the process;
- System management: from data found by the model developed, these are available through a computer interface;
- Industrial process: this block represents the electricity consuming process. It could, if necessary, be managed by the proposed system.

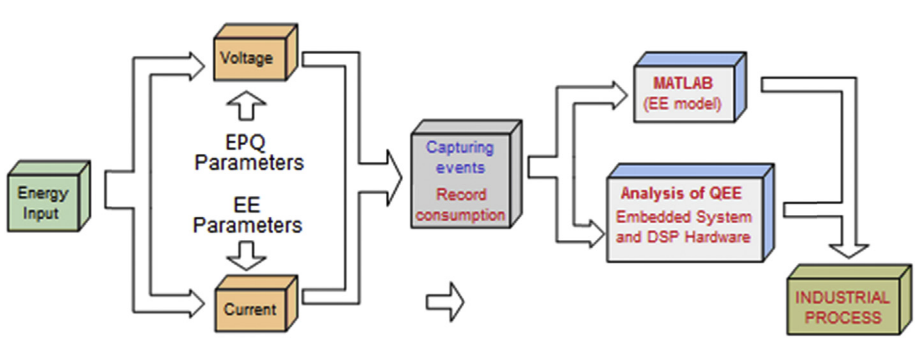

Fig. 4. Block diagram of the proposed system.

4.1. Embedded DWT multiresolution algorithm

The development of an algorithm for disturbance analysis by DWT is needed in this project as an alternative to the use of proprietary software (such as MATLAB, for example). Also, it is important to have the algorithm available for the embedded platform.

A practical way of performing the DWT Multiresolution Analysis is through the Mallat pyramidal algorithm, which consists in dividing the original signal. One part is the original signal smoothed and the other is a magnification of the fluctuations or "noises" of the analyzed signal. In Fig. 5 a, there is an outline of the Mallat pyramidal algorithm. Fig. 5 b illustrates the data stream used to obtain the wavelet coefficients using the Mallat algorithm to perform the MRA of a signal X composed of N samples. Note that the approximation and detail coefficients are neatly organized in an array of wavelet coefficients (C), with the value of their size depending on the number of signal samples and the selected decomposition level. Likewise, it creates another vector (L) corresponding to the sizes of each block of approximation and detail coefficients stored in C.

Thus, an algorithm was developed for disturbances analysis to execute in embedded applications, comprising the following features: (i) DWT Multiresolution Analysis and (ii) Haar mother wavelet. The use of the Haar mother wavelet corresponding to matrix multiplication is computed according to equation (3):

$$\begin{bmatrix} k_1 & k_2 \\ k_1 & k_2 \\ \ldots & \ldots \\ k_{n-1} & k_n \end{bmatrix} \times \begin{bmatrix} c_0 & c_0 \\ c_1 & -c_1 \end{bmatrix} = \begin{bmatrix} a_m & d_m \\ a_m & d_m \\ \ldots & \ldots \\ a_m & d_m \end{bmatrix} \tag{3}$$

on what $c_0 = \frac{\sqrt{2}}{2}$ e $c_1 = \frac{\sqrt{2}}{2}$ (Haar wavelet coefficients).

The flowchart shown in Fig. 6 models the implemented embedded algorithm. In this flowchart, the Haar function is applied to the signal S and two sub signals are generated. The signal $H(s)$ represents the wavelet coefficients resulted from the application of the Haar Wavelet Transform.

Based on this diagram, an algorithm was developed in C language for multiresolution analysis from the application of wavelet Haar's mother. The adoption of C programming language is due to its portability to any platform like a general purpose computer, an embedded system, and even a more basic microcontroller system. For the developed MRA algorithm validation, vectors of 64, 256 and 1024 positions were tested, being their decomposition occurred until the level 3. This validation made possible the obtained results comparison using both the embedded developed platform and Matlab simulation.

4.2. Study case scenario

The project validation consists on a power quality and energy demand monitoring process, performed in a tobacco beneficiation process. This scenario was chosen due to the high energy consumption presented in its processes, besides the large amount of power elements installed such as three-phase induction motors, boilers, reactors, among others.

The process wherein the monitoring system has been installed corresponds to the tobacco processing line and it is characterized by the following electrical charges:

- About 280 induction motors, mostly three-phase;
- Approximately 250 engines with direct start;
- Other 32 engines controlled by frequency inverters;
- A boiler for heated steam.

The process has a high power substation, and contains equipments for electrical energy transmission and distribution, as well as protection and control equipments. It is also responsible to decrease power from distribution company, usually 13,800 V used as working voltage for industrial equipments (380 VAC for phase-to-phase voltage). Considering the substation that feeds the tobacco processing line, this has seven fillister transformers connected in parallel to a bus, with a total power of 2800 kV A. Thus, it has a single voltage value for each phase, since all transformers are connected to the same electric potential, but there are distinct streams in each of the transformers, and each of these currents depends on the transformer power (Fig. 7).

The voltage reading on the transformer secondary was performed by the Current Transformer (CT) with transforming values between 1500 A and 5 A, with rated load from 0.6 to 5 V A. The accuracy of the system is about 2%, considering sensor sensitivity, linear range, A/D converter resolution, adopted measurement method and equipment used for calibration.

4.3. Acquisition hardware implementation

The signal acquisition is performed using current sensors and voltage sensors connected to the grid, corresponding to the step of

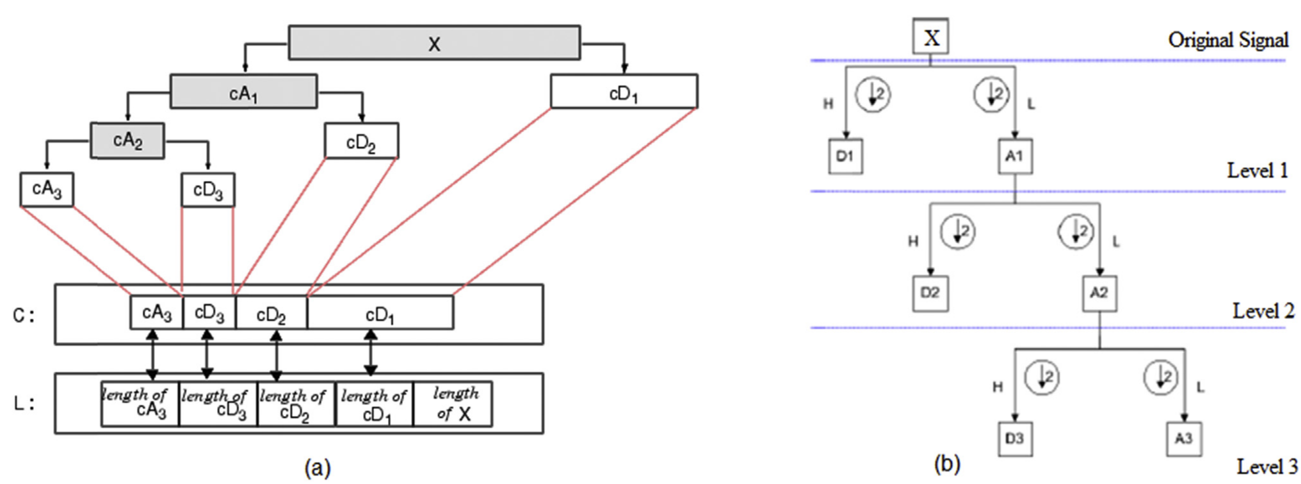

Fig. 5. Tree wavelet decomposition up to level 3. Source: [7,19].

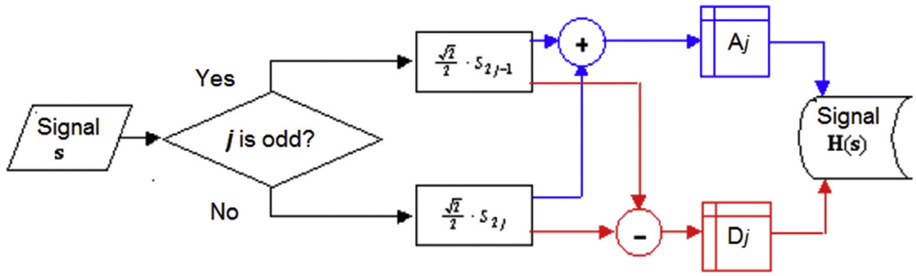

Fig. 6. Flow diagram of DWT. Source: [22].

signal input. This signal is conditioned to achieve specific value ranges and within the limits of the analog digital converter (ADC) used in digital signal processing (DSP). As output, it has a discretized signal, and thus it can be computationally manipulated.

The last stage of the system development is the integration of all devices already developed in this project. To start monitoring a three-phase production system, it is necessary at least an embedded system, a DSP platform, conditioning modules for TC signals, conditioning modules for TP signals, one power supply and one converter for communication between devices.

The experimental scenario involves data capturing from different points (multiple transformers). Thus it was necessary to install data acquisition systems on each of the points to capture the required information. Considering the amount of information collected in each equipment, it was decided to centralize the registry of such information on a single embedded system platform. For data communication and transference between devices it was chosen the standard RS485 protocol and data access can be done via Local Area Network (LAN), according to Fig. 8.

5. Results

5.1. Consumption analysis

This stage corresponded to the verification of power consumption behavior by the system and its efficiency. The data to be analyzed corresponds to the amount of current consumed in each phase of the transformers monitored by the data acquisition systems. The implemented system monitored the manufacturing process for about a month, and during this period, the company operated in a single round, with opening hours from 06:20 a.m. until 01:30 p.m.

The data collected in this period allowed the verification of the EPQ and trace the pattern of consumption (EE) for the process. For example, Fig. 9 shows the behavior of the energy demand (in terms of Current (A)) monitored and recorded during a production day. In this image it is possible to observe the similarity between the three curves, representing the demand of the three transformers installed at the substation.

5.2. Consumption model - EE

The consumption model can be taken from the comparison between the demand curves (Current (A) recorded during the study period. Fig. 10 illustrates these curves, which present the data consumption of 4 days (05/11 06/11, 07/11 and 08/11). In the picture one can see the similarity among all curves.

It is also possible to observe the behavior of the consumption curve, initially peaking around 6:30 a.m., and remaining constant until the end of the shift, around 02:00 p.m. Therefore, there is a demand of approximately 9 h, corresponding to the start time and end of the shift. Thus, any current event which exceeds these two limits by more than 1 min indicates a disturbance or abnormal operation.

5.3. Analysis of EPQ and EE patterns

The parameters of EPQ in the electric current also affect the EE standards and should be considered in all analysis and monitoring processes. In this paper, these disorders are analyzed using the values of average consumption, or RMS current values (I_{RMS}), for each phase of the transformers. These disorders correspond to

Fig. 7. Current measurement in a single transformer. The detail shows the installed CTs in vertical bus.

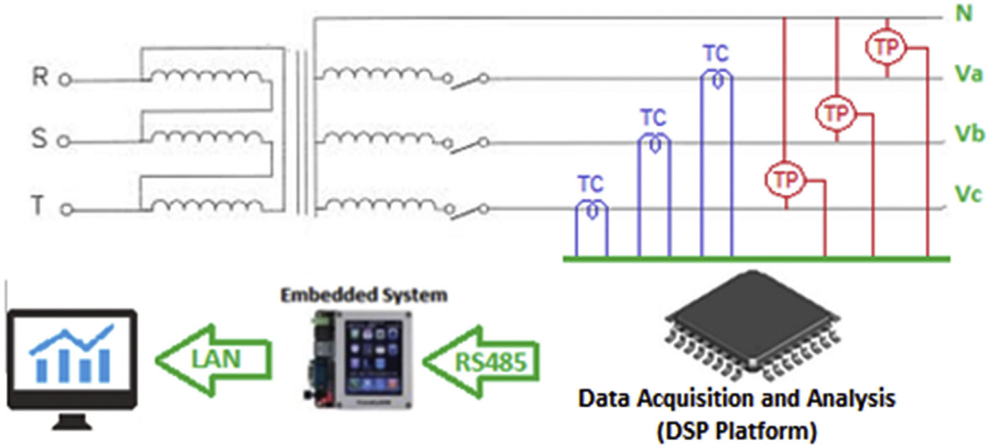

Fig. 8. Data acquisition and analysis system.

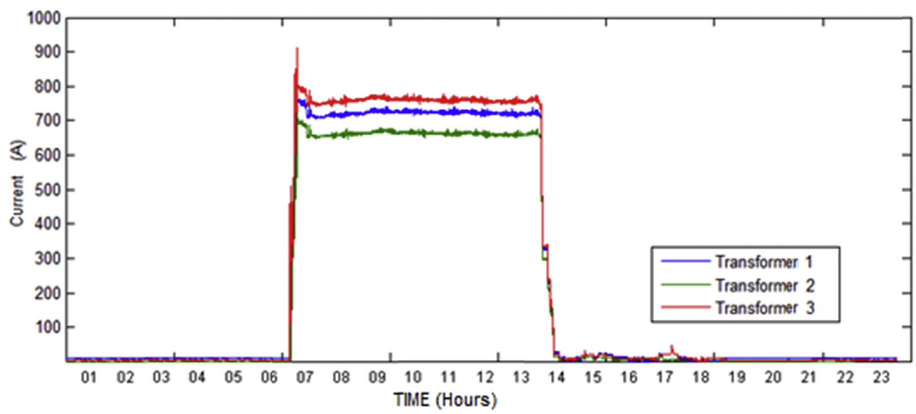

Fig. 9. Graphic with current IR of the three transformers.

abrupt events that occur instantaneously, changing the behavior and consumption curve of the system. One example is the activation of large loads such as induction motors for direct start generating current peaks several times the rated current. Likewise, the detachment of device also causes a sudden drop in current, or sags.

Fig. 11 shows the current consumption curve of the process during one shift (24 h). It is possible to observe the activation of the loads until the stabilization of the process, with current demand close to 800 Amps. However, there is a dip in current, caused by the equipment reset, occurred at 08:27:33 due to over-voltage.

Based on the consumption graphic on Fig. 11, the DWT is applied to locate the disturbance current. As a result there is the wavelet spectrum chart (bottom graphic on Fig. 10), corresponding to the noise present in the consumption curve after the DWT analysis using the Haar mother wavelet. For this analysis, the most important points of the graphic are the jumps of greater extent, indicating the times when there were abrupt changes in the consumption curve.

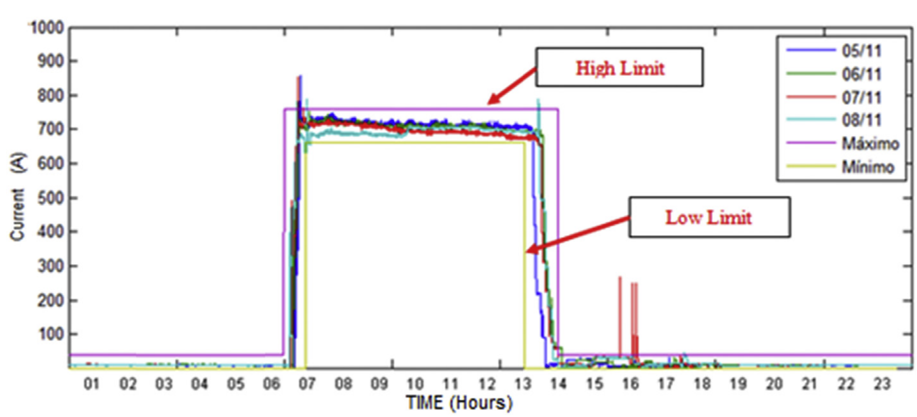

Fig. 10. Graphic for model consumption.

Fig. 11 illustrates the time required for process restoration, or the time required for the behavior of that consumption curve to fit to their model feature (around 800 Amps). In this period, which lasted about 40 min, workers were forced to stop their activities, and tobacco could not be processed.

The analysis of EPQ sinusoidal voltage disturbances data, collected by the embedded system (for single phase), can be seen in Fig. 12. In this figure it is possible to observe the change in the wavelet spectrum level at the time when the amplitude of the sine wave suddenly rises (high voltage short duration). This impulse response in the spectrum corresponds to the coefficients of detail when a disturbance is inserted in the original signal and is easily detected by the wavelet function due to its analytical characteristic in the time domain.

All information results captured in the factory floor were also validated by Matlab simulation. In this test, different mother wavelets were applied. However, the best results were obtained using db4, once Haar mother wavelet more efficient for effective voltage analysis.

6. Conclusion

With production costs getting higher, it is essential that companies monitor their energy expenses in order to become more competitive, and maximize their profits. However, the systems for monitoring and recording EPQ and EE available in the market still have high costs and are generally restricted to collecting data at a single point. Thus, this paper presented the results achieved with the development of a low cost embedded system targeted to the management of electricity in industrial processes. From the data collected, it was observed that the integration of different technologies associated to methodologies for capturing and signals analysis result in tools that can assist in monitoring and effective management of energy demands, thus contributing to a better Energy Efficiency in the industry sector.

The analysis of consumed demand (total demand by area, sector and hours) can provide a model of partial consumption. However it

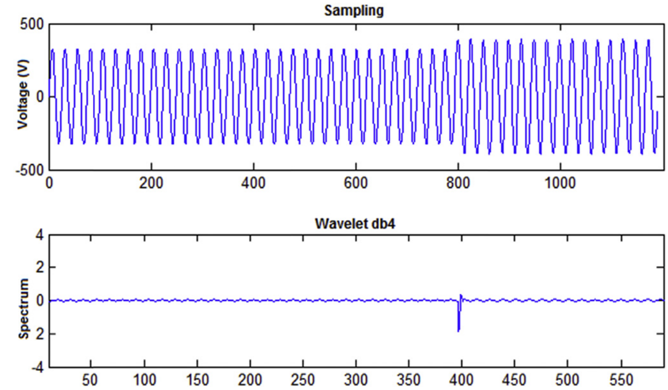

Fig. 12. High voltage analysis by DWT.

can illustrate the behavior of the electrical consumption of the industrial process. The algorithm developed to analyze demand offers, besides checking the demand consumed, does the profiling of consumption for each day or week, in order to estimate consumption peaks, thereby avoids the exceeding of the contracted demand.

Regarding the costs of the project, it can be considered that the system has an acceptable level as it requires a minimum of electronic elements such as power supply and data acquisition and conditioning boards, being the main task performed by the embedded platform. Since the embedded system provides an interface between the data collected and the end user, it has a cost value several times smaller than a general purpose computer. Furthermore, this type of system allows an accurate local evaluation of EPQ, making possible to transmit this analysis to a centralized management computer system. This system, successively, can make more qualified decisions, based on critical EPQ parameters measurements and analysis from distributed embedded monitoring devices. In fact, this concept is closed to the adoption of IIoT (Industrial Internet of Things) in modern industrial environments,

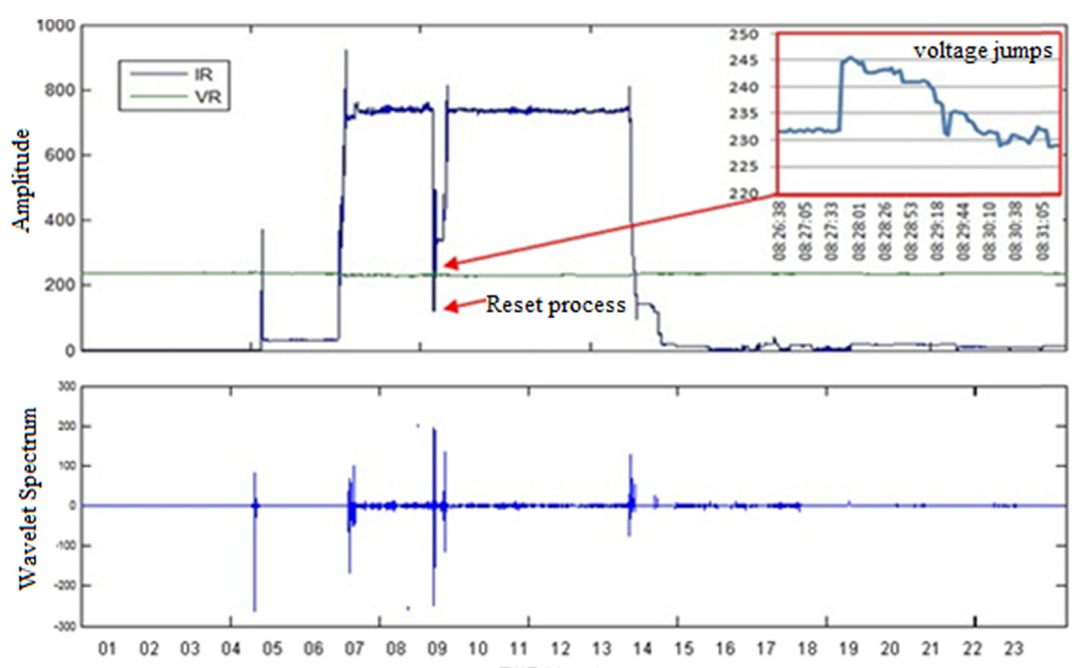

Fig. 11. Perturbation analysis and application of DWT Haar.

with small computers performing sensing and controlling on an entire process.

For future work, it is desired to improve the storage of this data system in a more robust database in order to make this information accessible to any management system. Additionally, it is been planned a system that executes analysis methodologies using parallel and distributed processing to extract profiles of energy consumption.

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